INCH-POUND

MIL-R-83580
23 November 1993

MILITARY SPECIFICATION

RESISTOR, FIXED, FILM, CHIP, ESTABLISHED RELIABILITY,
GENERAL SPECIFICATION FOR

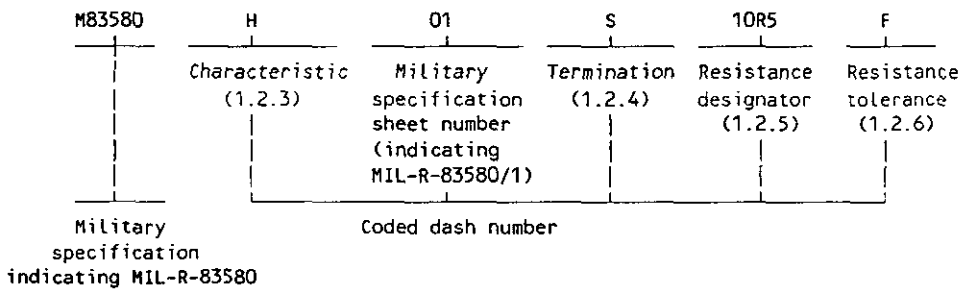
This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements for fixed, film, chip resistors primarily intended for incorporation into hybrid microelectronic circuits and surface mounting applications used in space, missile and other high reliability applications. The prerequisite to qualifying to this specification shall be based on current qualification to 'R' failure rate level or better for detail specifications of similar MIL-R-55342 resistor styles. This product shall be capable of meeting all requirements of MIL-R-55342, in addition to the requirements specified herein.

1.2 Classification.

1.2.1 Part or Identifying Number (PIN). Resistors specified herein (see 3.1) shall be identified by a PIN which shall consist of the basic number of the military specification sheet and a dash number. Each military specification sheet covers a different resistor style. The dash number shall be coded to provide information concerning the resistor characteristic, resistance value, resistance tolerance, and terminal material. The PIN shall be in the following form with the dash number derived as indicated:



NOTE: MIL-R-83580/7 uses a "D" prefix (e.g. D83580H07XXXXXX).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: SMC/SDFP, P.O. Box 92960, Worldway Postal Center, Los Angeles, CA 90009-2960, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of the document or by letter.

AMSC N/A

FSC 5905

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1.2.2 Style. The style (see 3.1) is identified by the three-letter symbol "RSM" followed by a four digit number; the letters identify fixed, film, resistors in a flat chip configuration.

1.2.3 Characteristic. The characteristic is identified by the single letter H, K, M, and E in accordance with table I.

TABLE I. Characteristics.

		Symbol			
		E	H	K	M
Resistance temperture coefficient (see 3.14)	Parts/million/°C	±25	±50	±100	±300
Maximum ambient temperature at full wattage (see 3.5)		70°C	70°C	70°C	70°C
Maximum ambient temperature at zero wattage (see 3.5)		150°C	150°C	150°C	150°C
Maximum percent change in resistance (0.01 ohm additional allowed for measurement error)	Thermal shock (see 3.8)	±.1%	±.25%	±.5%	±.5%
	Low temperature operation (see 3.9)	±.1%	±.25%	±.25%	±.5%
	Short time overload (see 3.10)	±.1%	±.1%	±.25%	±.5%
Maximum percent change in resistance (0.01 ohm additional allowed for measurement error)	Power conditioning (see 3.11)	±.2%	±.25%	±.5%	±.5%
	Resistance to bonding exposure (see 3.12)	±.2%	±.25%	±.25%	±.25%
	Moisture resistance (see 3.13)	±.2%	±.4%	±.5%	±.5%
Resistance tolerance ± percent (see table V)		.1 .5 1.0 2.0	.1 .5 1.0 2.0 5.0	.1 .5 1.0 2.0 5.0	1.0 2.0 5.0

1.2.4 Termination material. The termination material designation shall be in accordance with table II.

1.2.5 Resistance. The nominal resistance expressed in ohms is identified by four characters, the first three characters shall be the significant figures and the fourth place shall be the number of zeros to follow. The letter 'R' shall be used to signify the decimal point and all digits are significant. Examples of resistance designators are as specified in table III. Table IV lists standard decade values for available resistors values in corresponding tolerances.

1.2.6 Resistance tolerance. The resistance tolerance is as identified by a single letter in accordance with table V.

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TABLE II. Termination material.

Type	Material	Termination area	Code letters
Solderable 1/ Bondable Weldable	Gold	Wraparound 2/ One surface Bonding pads	G W 3/ P
Solderable 1/	Base metallization barrier metal, solder coated	Wraparound 2/	B
	Pretinned	Wraparound 2/ One surface	S 3/
Bondable weldable	Platinum/gold	Wraparound One surface	U T 3/
Bondable	Palladium/silver	Wraparound One surface	C D 3/
	Platinum/silver	Wraparound One surface	K M 3/

1/ Bondable termination shall meet the solderability test. Solderable termination shall be pretinned for solder reflow operation and shall meet the solderability test.

2/ On wraparound termination, the pretinning shall be, as a minimum, on at least two sides and only those surfaces must meet the solderability test.

3/ See 3.4.2.2 and 6.4.4.

TABLE III. Designator of resistance values for 10 to 100 decade.

Resistance designator	Resistance ohms
1R00 to 9R88 incl.	1.0 to 9.88
10R0 to 98R8 incl.	10.0 to 98.8
1000 to 9880 incl.	100 to 988
1001 to 9881 incl.	1,000 to 9,880
1002 to 9882 incl.	10,000 to 98,800
1003 to 9883 incl.	100,000 to 988,000
1004 to 9884 incl.	1,000,000 to 9,880,000
1005	10,000,000

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TABLE IV. Standard resistance values for the 10 to 100 decade.

0.1	1.0	2.0 5.0	0.1	1.0	2.0 5.0	0.1	1.0	2.0 5.0	0.1	1.0	2.0 5.0
10.00	10.00	10.00	20.00	20.00	20.00	37.40	37.40		69.80	69.80	
10.10			20.30			37.90			70.60		
10.20	10.20		20.50	20.50		38.30	38.30		71.50	71.50	
10.40			20.80			38.80			72.30		
10.50	10.50		21.00	21.00				39.00	73.20	73.20	
10.60			21.30			39.20	39.20		74.10		
10.70	10.70		21.50	21.50		39.70			75.00	75.00	
10.90			21.80			40.20	40.20		75.90		
11.00	11.00	11.00			22.00	40.70			76.80	76.80	
11.10			22.10	22.10		41.20	41.20		77.70		75.00
11.30	11.30		22.30			41.70			78.70	78.70	
11.40			22.60	22.60		42.20	42.20		79.60		
11.50	11.50		22.90			42.70			80.60	80.60	
11.70			23.20	23.20				43.00	81.60		
11.80	11.80		23.40			43.20	43.20				
12.00		12.00	23.70	23.70		43.70			82.50	82.50	82.00
12.10	12.10		24.00		24.00	44.20	44.20		83.50		
12.30			24.30	24.30		44.80			84.50	84.50	
12.40	12.40		24.60			45.30	45.30		85.60		
12.60			24.90	24.90		45.90			86.60	86.60	
12.70	12.70		25.20			46.40	46.40		87.60		
12.90			25.50	25.50		47.00		47.00	88.70	88.70	
13.00	13.00	13.00	25.80			47.50	47.50		89.80		
13.20			26.10	26.10		48.10			90.90	90.90	
13.30	13.30		26.40			48.70	48.70				91.00
13.50			26.70	26.70		49.30			92.00		
13.70	13.70				27.00	49.90	49.90		93.10	93.10	
13.80			27.10			50.50			94.20		
14.00	14.00		27.40	27.40				51.00	95.30	95.30	
14.20			27.70			51.10	51.10		96.50		
14.30	14.30		28.00	28.00		51.70			97.60	97.60	
14.50			28.40			52.30	52.30		98.80		
14.70	14.70		28.70	28.70		53.00					
14.90			29.10			53.60	53.60				
15.00	15.00	15.00	29.40	29.40		54.20					
15.20			29.80			54.90	54.90				
15.40	15.40				30.00	55.60					
15.60			30.10	30.10				56.00			
15.80	15.80		30.50			56.20	56.20				
16.00		16.00	30.90	30.90		56.90					
16.20	16.20		31.20			57.60	57.60				
16.40			31.60	31.60		58.30					
16.50	16.50		32.00			59.00	59.00				
16.70			32.40	32.40		59.70					
16.90	16.90		32.80			60.40	60.40				
17.20					33.00	61.20					
17.40	17.40		33.20	33.20		61.90	61.90				
17.60			33.60					62.00			
17.80	17.80		34.00	34.00		62.60					
18.00		18.00	34.40			63.40	63.40				
18.20	18.20		34.80	34.80		64.20					
18.40			35.20			64.90	64.90				
18.70	18.70		35.70	35.70		65.70					
18.90					36.00	66.50	66.50				
19.10	19.10		36.10			67.30					
19.30			36.50	36.50				68.00			
19.60	19.60		37.00			68.10	68.10				
19.80						69.00					

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TABLE V. Resistance tolerance.

Resistance tolerance (±%)	Designator
0.1	B
1.0	F
2.0	G
5.0	J

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

QQ-S-571 - Solder; Tin Alloy; Lead-Tin Alloy; and Lead Alloy.
MMM-A-1931 - Adhesive, Epoxy, Silver Filled, Conductive.

MILITARY

MIL-R-39032 - Resistors, Packaging of.
MIL-I-46058 - Insulating, Compound, Electrical (For coating printed circuit assembles).
MIL-R-55342 - Resistors, Fixed, Film, Chip, Established Reliability, General Specification for.

(See supplement 1 for list of associated detail specifications).

STANDARDS

MILITARY

MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.
MIL-STD-790 - Reliability Assurance Program for Electronic Part Specifications.
MIL-STD-1285 - Marking of Electrical and Electronic Parts.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, Robins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications from a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

NATIONAL AERONAUTICS AND SPACE ADMINISTRATION (NASA)

NASA 1124 - Outgassing Data for Selecting Spacecraft Materials.

(Copies of NASA 1124 publication are available from NASA/Parts Project Office (NPPO), NASA Goddard Space Flight Center, Code 310.A, Greenbelt, MD 20771.)

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2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the document cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E595 - Materials from Outgassing in a Vacuum Environment, Total Mass Loss and Collected Volatile Condensable, Standard Test Method for.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Non-Government standards and other publications are normally available from the organizations that prepare of distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated detail requirements. The individual item requirements shall be as specified herein and in accordance with the applicable detail specification. In the event of any conflict between the requirements of this specification and the associated detail specification, the latter shall govern (see 6.2).

3.2 Qualification. Resistors furnished under this specification shall be products which are authorized by the qualifying activity for listing on both qualified products list at the time of the award of contract (see 4.4 and 6.3) for MIL-R-55342 (with minimum failure rate 'R') and herein. The manufacturer shall obtain certification from the qualifying activity that the reliability assurance requirements of 4.1.2 have been met and are being maintained. Unless acquired from a manufacturer and approved for listing, on the qualified products list, parts furnished under this specification shall not be considered as having met the requirements of this specification. Distributors shall not be used for distribution of parts. In addition, as a prerequisite for qualification, the manufacturer shall demonstrate to the qualifying activity that the requirements for outgassing have been met (see 3.19).

3.2.1 Reliability and baseline documentation. Resistors supplied to this specification shall conform to the requirements of MIL-STD-790 and the requirements specified herein. Prior to granting of qualification approval, a baseline document shall be prepared by the manufacturer and submitted to the qualifying activity for approval.

3.3 Material. The material shall be as specified herein and in accordance with the baseline document (see 3.2.1). In addition, materials used in the manufacturing of the resistor shall meet the outgassing requirements (see 3.19).

3.4 Design and construction. Resistors shall be of the design, construction, and physical dimensions specified (see 3.1). Each resistor shall consist of a film type resistance element connected to metallized contact areas of terminations which form the electrical contacts of the resistor. Film thickness shall be greater than 350 angstroms.

3.4.1 Passivation. Unless passivation is inherent in the device, the construction shall be passivated with a coating over the resistance element to protect the element from direct exposure to environment or handling contamination. Does not apply to trim cut.

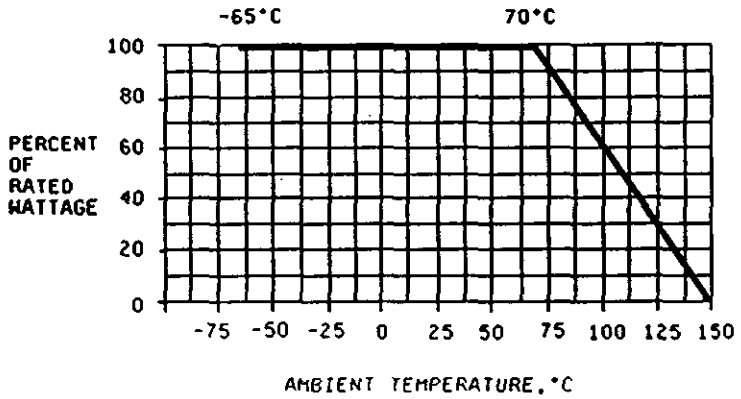
3.4.2 Termination. Connection to the resistor element shall be made via metallized termination areas. Unless otherwise specified, both terminations (areas) shall be available on one surface of the body of the resistor chip. See 1.2.4 for termination codes.

3.4.2.1 Barrier metallization. The barrier metallization for the B and G terminations (base metallization barrier metal, solder coated) shall be nickel. The metallization shall be a minimum of 50 microinches.

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3.4.2.2 Tin plated finishes. Use of tin plating is prohibited as a final finish and as an undercoat effective from the date of this specification (6.4.5). Use of tin-lead (Sn-Pb) finishes are acceptable provided that the minimum lead content is 3 percent.

3.5 Power rating. The resistors shall have a power rating based on continuous full-load operation at an ambient temperature of 70°C. For operation at temperatures in excess of 70°C, the load shall be derated in accordance with figure 1.



NOTE: This curve indicates the percentage of nominal wattage to be applied at temperature higher than 70°C. This curve applies only to units mounted on a substrate (see 4.7.1.3); however, at no time shall the applied voltage exceed the maximum for each style (see 3.1).

FIGURE 1. Derating curve for high ambient temperatures.

3.6 Voltage rating. Resistors shall have a rated dc continuous working voltage or an approximate sine-wave root-mean-square (rms) continuous working voltage corresponding to the wattage (power) rating, as determined from the following formula:

$$E=\sqrt{(PR)}$$

Where; E = continuous rated dc or rms working voltage in volts
P = rated wattage in watts
R = nominal resistance in ohms

In no case shall the rated voltage be greater than the applicable maximum voltage (see 3.1).

3.7 DC resistance. When resistors are tested as specified in 4.7.2, the dc resistance shall be within the specified tolerance of the nominal resistance (see 1.2.5).

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3.8 Thermal shock. When resistors are tested as specified in 4.7.3, there shall be no evidence of mechanical damage; the combined change in resistance for thermal shock and power conditioning during group A inspection shall not exceed ±(0.25 percent +0.01 ohm) for characteristic H. The change in resistance for group B shall not exceed ±(0.25 percent +0.01 ohm) for characteristic H, ±(0.50 percent +0.01 ohm) for characteristic K and M, and ±(0.1 percent +0.01 ohm) for characteristic E.

3.9 Low temperature operation. When resistors are tested as specified in 4.7.4, there shall be no evidence of mechanical damage; the change in resistance between the initial and the final measurement at 25°C ±5°C, shall not exceed ±(0.25 percent +0.01 ohm) for characteristic H, ±(0.25 percent +0.01 ohm) for characteristic K, ±(0.5 percent +0.01 ohm) for characteristic M, and ±(0.1 percent +0.01 ohm) for characteristic E.

3.10 Short time overload. When resistors are tested as specified in 4.7.5, there shall be no evidence of arching, burning, or charring. The change in resistance shall not exceed ±(0.1 percent + 0.01 ohm) for characteristic H, ±(0.25 percent +0.01 ohm) for characteristic K, ±(0.5 percent +0.01 ohm) for characteristic M, and ±(0.1 percent +0.01 ohm) for characteristic E.

3.11 Power conditioning. When resistors are tested as specified in 4.7.6, there shall be no evidence of mechanical damage; the combine change in resistance, for thermal shock and power conditioning shall not exceed ±(0.25 percent +0.01 ohm) for characteristic H, ±(0.5 percent +0.01 ohm) for characteristic K and characteristic M, and ±(0.2 percent +0.01 ohm) for characteristic E.

3.12 Resistance to bonding exposure. When resistors are tested as specified in 4.7.7, there shall be no evidence of mechanical damage, no demetallization or leaching and the change in resistance shall not exceed ±(0.25 percent +0.01 ohm) for characteristic H, K, and M, and ±(0.2 percent +0.01 ohm) for characteristic E.

3.13 Moisture resistance. When resistors are tested as specified in 4.7.8, there shall be no evidence of mechanical damage; the change in resistance between the initial and final measurements shall not exceed ±(0.4 percent +0.01 ohm) for characteristic H, ±(0.5 percent +0.01 ohm) for characteristic K and M, and ±(0.2 percent +0.01 ohm) for characteristic E.

3.14 Resistance temperature coefficient. When resistors are tested as specified in 4.7.9, the resistance-temperature coefficient, at each of the temperatures specified in table XI referred to room temperature, shall not exceed ±.0050 percent per °C (50 ppm) for characteristic H, ±.0100 percent per °C (100 ppm) for characteristic K, ±.0300 percent per °C (300 ppm) for characteristic M, and ±.0025 percent per °C (25 ppm) for characteristic E.

3.15 Solderability (applicable to B, G, and S terminations). When resistors are tested as specified in 4.7.10, the immersed metalized surface shall be at least 95 percent covered with a new clean smooth coating and shall exhibit no demetallization or leaching of the terminal areas. The remaining 5 percent may contain only small pinholes or rough spots, and shall not be concentrated in one area. In case of a dispute, the percentage of coverage with pinholes or rough spots shall be determined by actual measurement of these areas, as compared to the total area.

3.16 Adhesion. When resistors are tested as specified 4.7.11, there shall be no evidence of mechanical damage.

3.17 Visual inspection. Resistors shall be inspected to verify that the materials, design, construction, physical dimensions, marking, and workmanship are in accordance with the applicable requirements (see 3.3, 3.4, 3.4.1 to 3.4.3, 3.17.1 to 3.17.5d, 3.21, and figures 2 through 13). The examinations shall be performed at 30X minimum magnification unless otherwise specified.

3.17.1 Conductor metallization defects. Any resistor which exhibits the following defects in the active circuit metallization shall not be acceptable. The active circuit metallization is all metal or any other material used for interconnection except metalized scribe lines, unused bonding pads and identification marking.

3.17.1.1 Conductor metallization scratches. A scratch is defined as any tearing defect that disturbs the original surface of the metallization.

a. Any scratch in metallization through which the underlying resistor material also appears to be scratched.

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b. Any scratch in the interconnecting metallization which exposes resistive material substrate or oxide anywhere along its length and reduces the width of the connecting strip to less than 50 percent of its original width.

3.17.1.2 Conductor metallization voids. A void is defined as any region in the interconnecting metallization where underlying resistive material substrate or oxide is visible which is not caused by a scratch. Any void in the interconnecting metallization which leaves less than 50 percent of the original width undisturbed (see figure 2).

3.17.1.3 Conductor metallization adherence. Metallization adherence is defined as any evidence of metallization lifting, peeling, or blistering (see figure 6).

3.17.1.4 Conductor metallization probing. Probing marks on the interconnecting metallization other than bonding pads that violate the scratch or void criteria.

3.17.1.5 Conductor metallization bridging (see figure 2).

a. Bridged metallization defect that reduces the distance between any two metallization areas to less than 50 percent or 0.3 mils, whichever is greater.

b. Bridging between metallization and resistor pattern not intended by design that reduces the distance between the two to less than 0.1 mils.

3.17.1.6 Conductor metallization missing.

a. More than 5 percent of the metallization area missing on a bonding pad (see figure 2).

b. Any shorting bar or metal pad is missing (see figures 2 and 9).

3.17.1.7 Conductor metallization alignment. Metallization alignment defined as any misalignment between resistor and metallization such that more than 0.5 mil of resistor on a side is exposed (see figure 3).

3.17.1.8 Conductor metallization corrosion. Metallization corrosion is defined as any evidence of localized heavy stains, metallization corrosion, discoloration or mottled metallization (see figure 8).

3.17.1.9 Excessive metallization. Excessive buildup of termination on any surface material greater than .003 inch (0.08 mm) and .008 (0.20 mm) for solderable and bondable terminations respectively (see figure 5).

3.17.2 Resistor defects. Any resistor which exhibits the following defects in the active area shall not be acceptable. The active area of a resistor is that part of a resistor pattern which remains in series connection between resistor terminals and is not shorted by metallization and the resistor element is not more than two rungs past laser trimmed link.

3.17.2.1 Resistor scratches (see figure 4).

a. Any scratch within the active resistor area.

b. A scratch will normally be of considerable length, 1 mil or longer and on fine lines, and will extend continuously on adjacent lines. A scratch must be indicated by either disturbance of the surface resistor or extraneous, shiny material remaining from the instrument causing the scratch.

3.17.2.2 Resistor voids (see figures 2 and 10).

a. Any void or neckdown in the active resistor which reduces the width of the area by more than 25 percent of the original area, except calibration cuts which shall not exceed 50 percent of the width of the resistors.

b. Any void in the resistor element at the termination.

3.17.2.3 Resistor adherence. Resistor adherence is defined as any evidence of resistor film lifting, peeling, or blistering (see figure 6.)

3.17.2.4 Probe marks. Probe marks are defined as any probe marks on the active resistor area.

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3.17.2.5 Resistor material corrosion.

a. Any evidence of localized heavy stains or corrosion of resistor material in the active resistor area (see figure 8).

b. Discoloration of tantalum based resistors due to thermal stabilization is not a cause for rejection.

3.17.2.6 Resistor bridging defects.

a. Resistor bridging caused by a photolithographic defect that forms a conductive bridge across two or more adjacent active resistance areas where width of bridge path is less than one half the smallest line width in the pattern.

b. Partial bridging defect that reduces the distance between adjacent active resistance areas to less than 0.1 mil or 50 percent of the design separation whichever is less when caused by smears, photolithorgraphic defects or other causes except that for a partial bridge within the lines and spacing of 0.1 mil design width, visual separation (evidence at 400X max.) is sufficient for acceptance, and that apparent bridging when caused by laser trim slag in a nonlaser trim area is acceptable as long as the resistor material under the apparent bridge is passivated.

3.17.2.7 Laser trim faults. Any resistor which exhibits the following defects in the active area shall not be acceptable.

a. Uncut material remaining after a laser cut due to "skipping" of laser beam (see figure 7).

b. If laser cut is not straight lined, the narrowest remaining width must be equal to or greater than the width of the narrowest line within the same resistor pattern.

c. Oxide voids, cracking, or similar damage caused to the underlying material by laser beam where such damage touches active interconnects, resistor area or both.

d. For a resistor pattern with links designed for incremental laser trim, where edge cuts touches the active resistor area.

e. Any "scorching" of the resistor areas by direct laser beam or spurious reflections caused by optics of the system.

f. Discoloration of tantalum based resistors in and around laser kerf is not cause for rejection.

g. For a cut in the resistor pattern which reduces a link or ladder width by more than 50 percent of the original width.

h. Usable resistance width is less than 50 percent of active resistor element (figure 10).

3.17.3 Oxide void. Any resistor which exhibits an oxide void shall not be acceptable. Any oxide void is a fault in the oxide evidenced by localized double or triple colored fringes at the edges of the defect at 100X max.

a. Oxide void that allowed bridging between any two resistors, metal areas or both, not intended by design.

b. Any evidence of oxide void under metallization on resistor geometry.

3.17.4 Scribing and substrate. The following defects shall be cause for rejection:

a. Less than 0.5 mil oxide visible between active metallization and edge of substrate. Excluded from this are any inactive metallization lines (if applicable).

b. Any chip or crack in the active resistor, metal area, or both.

c. Any crack that exceeds 3.0 mil in length or comes closer than 1.0 mil to an active resistor area on the substrate (see figure 11).

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d. Any crack in a substrate that exceeds 1.0 mil in length and points toward the active resistor area.

e. A substrate, when scribed and broken, is left with an attached portion of an adjacent substrate which contains metallization or resistor material.

f. A crack or chip in the backside of a substrate that leaves less than 75 percent of area or, a crack or chip under bonding pad.

g. Any crack not originating at a substrate edge is cause for rejection (see figure 11).

3.17.5 Foreign material. Conductive foreign material is defined as any substance which appears opaque under all conditions of lighting and magnification used in routine visual inspection. Material is considered attached when it can not be removed with a nominal gas blow (20 psig) of dry nitrogen or air.

a. Attached conductive foreign material which reduces the separation between metallization areas to less than 0.1 mil is cause for rejection.

b. Attached conductive foreign material which reduces the separation between metallization and resistor areas to less than 0.1 mil is cause for rejection.

c. More than seven attached nonconductive or conductive foreign material that are not rejectable by the criteria of A and B above shall be cause for rejection.

d. Ink residue on chip from tester shall be cause for rejection.

3.18 Destructive physical analysis. When examined as specified in 4.7.12 resistors shall meet the requirements in appendix B of this document.

3.19 Outgassing. When examined as specified in 4.7.13, the samples shall meet the following requirements:

a. Total Mass Loss (TML) shall not exceed 1 percent.

b. Volatile Condensable Material (VCM) shall not exceed .1 percent.

3.19.1 Outgassing test data. Data listed in NASA Publication 1124 may be used in lieu of actual test data for applicable materials.

3.20 Minimum marking legibility.

3.20.1 Laser marked. When tested as specified in 4.7.1.1 the marking shall remain legible.

3.20.2 Color dot/screen printing/stamp. When tested as specified in 4.7.1.2 the marking or color coding shall remain legible.

3.21 Marking. Marking of individual chip resistors is not required but is permitted. A noncorrosive label containing the following information shall be inserted in each unit package. PIN designation, "JAN" marking, date code, source code, and manufacturer's production lot code (see 6.2). Date and source code shall be in accordance with MIL-STD-1285. The following is an example of the complete marking:

12345	-	CAGE code
7343J	-	Date code and JAN marking
M83580H01	-	PIN
S10R5	-	Part number
XXXXX	-	Manufacturer's production lot code

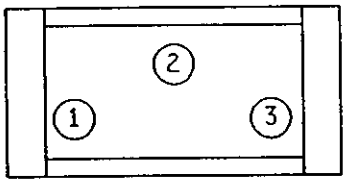
The date codes shall be the date of the final assembly operation for the production lot as defined in 6.2. The common manufacturing record shall include the same date code as that placed on the parts covered by the record. At the option of the manufacturer, resistors may be marked (i.e., laser, color dot, etc). The marking process must be approved by the qualifing activity based on testing specified herein. The marking shall remain legible after all tests.

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3.21.1 Color dot. (As an option for the manufacturer.)

3.21.1.1 3 dot scheme. (Designates 2 percent, 5 percent and 10 percent resistance values.)

1st dot - 1st significant figure.
2nd dot - 2nd significant figure.
3rd dot - Multiplier.



EX. 1 330 kΩ -

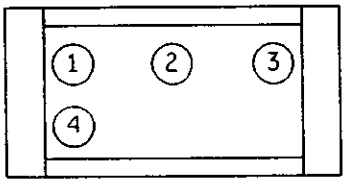
orange
orange yellow

EX. 2 12Ω -

red
brown black

3.21.1.2 4 dot scheme. (Designates 1 percent resistance values.)

1st dot - 1st significant figure.
2nd dot - 2nd significant figure.
3rd dot - 3rd significant figure.
4th dot - Multiplier (fractional multiplier are the same as described in the 3 dot scheme. ref. MIL-STD-1285).



EX. 1 30.1 kΩ -

orange - black - brown
red

EX. 2 18.2Ω -

brown - gray - red
gold (0.1)

3.21.2 Laser marking/screen printing/stamping.

		example of code
3.21.2.1 3 character.	ex. 330 kΩ	334
	12Ω	120
3.21.2.2 4 character.	ex. 30.1 kΩ	3012
	18.2Ω	18R2

3.21.3 JAN and J marking. The United States Government has adopted, and is exercising legitimate control over the certification marks "JAN" and "J", respectively, to indicate that items so marked or identified are manufactured to, and meet all the requirements of military specifications. Accordingly, items acquired to, and meeting all of the criteria specified herein and in applicable specifications shall bear the certification mark "JAN" except that items too small to bear the certification mark "JAN" shall bear the letter "J". The "JAN" or "J" shall be placed immediately before the PIN except that if such location would place a hardship on the manufacturer in connection with such marking, the "JAN" or "J" may be located on the first line above or below the PIN. Items furnished under contracts or orders which either permit or require deviation from the conditions or requirements specified herein or in applicable specifications shall not bear "JAN" or "J". In the event an item fails to meet the requirements of this specification and the applicable specification sheets or associated detail specifications, the manufacturer shall remove the "JAN" or the "J" from the sample tested and also from all items represented by the sample. The "JAN"' or "J" certification mark shall not be used on products acquired to contractor drawings or specifications. The United States Government has obtained Certificate of Registration Number 504,860 for the certification mark "JAN".

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3.21.4 Supplying to lower resistance temperature characteristics and or lower resistance tolerances. Parts qualified to lower resistance temperature characteristics and/or lower resistance tolerance levels are substitutable for higher resistance temperature or higher tolerance levels as long as the label reflects the correct resistance temperature characteristic and/or resistance tolerance (see tables VI and VII). Parts that are physically marked shall not be remarked unless specified in the contract or purchase order (see 6.2).

TABLE VI. Resistance tolerance substitution.

Resistance tolerance	Resistance tolerance substitution
B	B, D, F, G, J, K
D	D, F, G, J, K
F	F, G, J, K
G	G, J, K
J	J, K
K	K

TABLE VII. Characteristic substitution.

Characteristic	Characteristic substitution
Y	Y, E, H
E	E, H
H	H
K	K, M
M	M

3.22 Workmanship. Resistors shall be processed in such a manner as to be uniform in quality and shall meet the requirements of 3.1, 3.3, 3.4, 3.4.1 through 3.4.2.1, 3.21 and 3.21.1, as applicable, and be free from other defects that will affect life, serviceability, or appearance, and shall pass the visual inspection as specified on figures 2 through 13.

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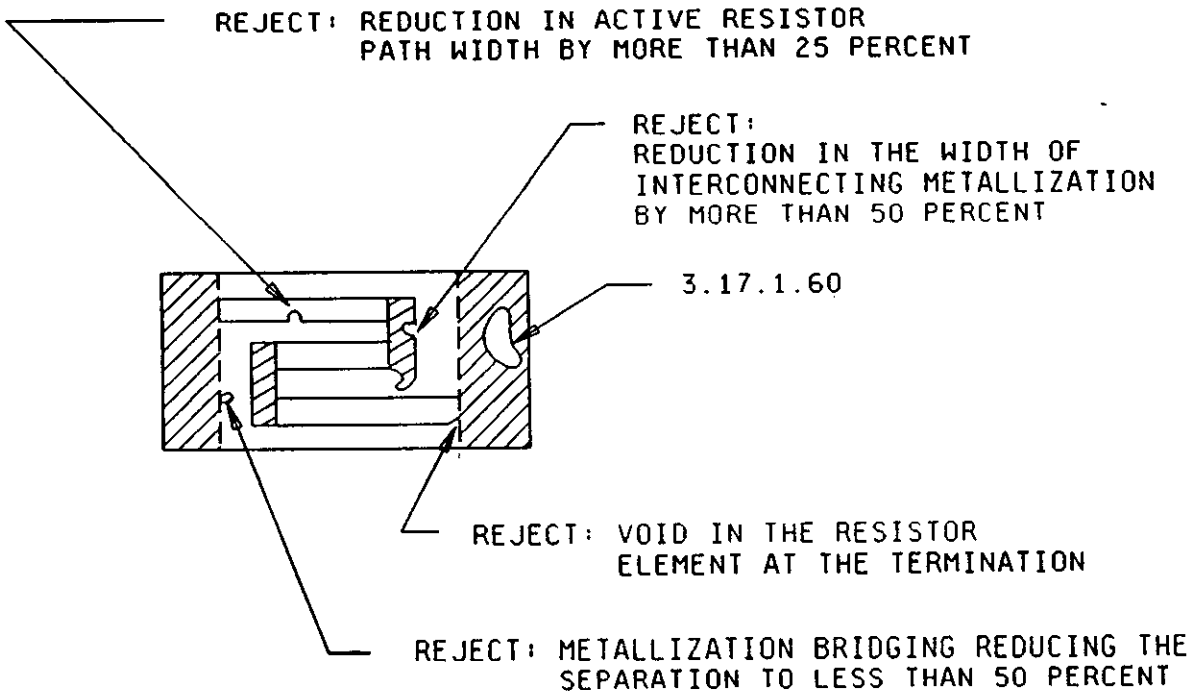
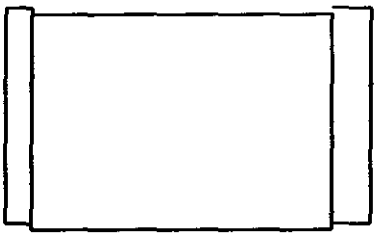


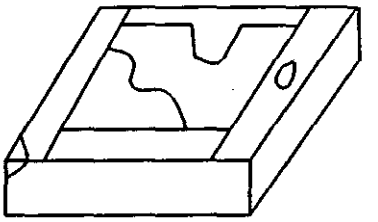
FIGURE 2. Metallization voids, bridging, missing metallization, and resistor voids.

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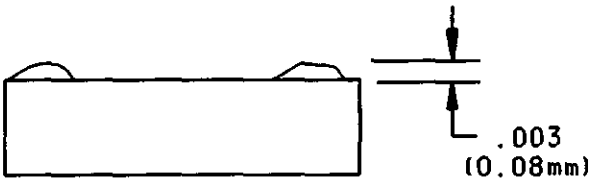
REJECT: ANY MISALIGNMENT BETWEEN RESISTOR AND METALLIZATION.

FIGURE 3. Metallization misalignment.



REJECT: ANY METALLIZATION VOIDS OR SCRATCHES ON RESISTANCE AREA

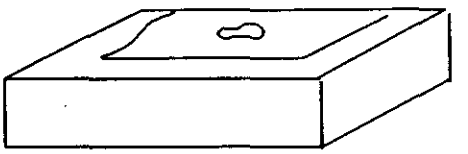
FIGURE 4. Metallization voids or scratches.



REJECT: EXCESSIVE BUILD UP OF TERMINATION ON ANY SURFACE MATERIAL GREATER THAN .003 INCH (0.08mm) AND .008 INCH (0.20mm) FOR SOLDERABLE AND BONDABLE LEADS, RESPECTIVELY.

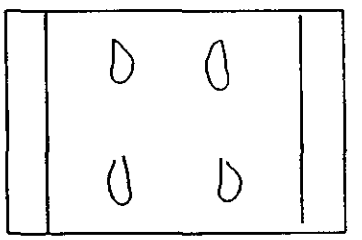
FIGURE 5. Excessive metallization.

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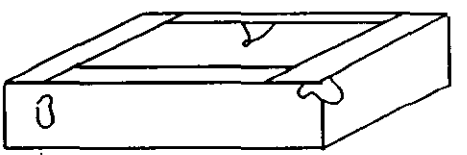
REJECT: ANY EVIDENCE OF PEELING, LIFTING OR BLISTERING.

FIGURE 6. Peeling lifting or blistering metallization.



REJECT: ANY LASER TRIM FAULTS (UNCUT MATERIAL REMAINING AFTER A LASER CUT DUE TO SKIPPING OF LASER BEAM, IF LASER CUT IS NOT STRAIGHT).

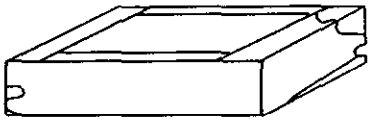
FIGURE 7. Laser trim faults.



REJECT: METALLIZATION CORROSION (LOCALIZED STAINS, CORROSION, DISCOLORIZATION, MOTTED METALLIZATION).

FIGURE 8. Metallization corrosion.

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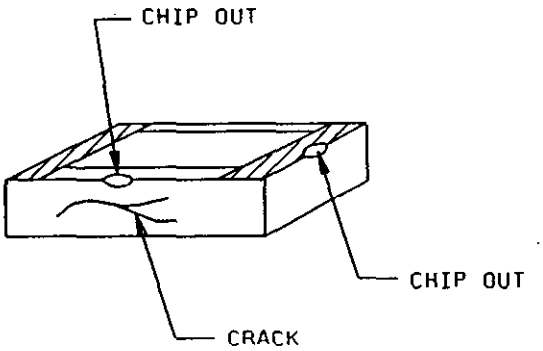
REJECT: MISSING BONDING PAD (ANY SHORTING BAR OR METAL IS MISSING UNLESS BY DESIGN).

FIGURE 9. Missing metallization.



REJECT: VOID NICK, BUBBLE OR CUT IN ACTIVE RESISTOR AREA GREATER THAN 25 PERCENT OF THE RESISTOR WIDTH EXCEPT FOR A CALIBRATION CUT AFTER CALIBRATION. THE USABLE RESISTANCE WIDTH FOR CHARACTERISTICS E,H,K AND M SHALL BE NOT LESS THAN 50 PERCENT OF THE ACTIVE RESISTOR ELEMENT

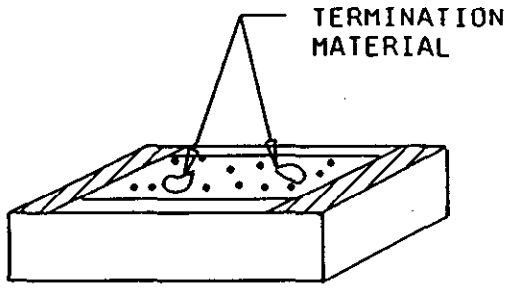
FIGURE 10. Voids, nicks, bubble or cuts.



REJECT: ANY CRACK, CHIP OUT OR VOID IN THE SUBSTRATE GREATER THAN .003 INCH (0.08mm) IN ANY ONE DIRECTION.

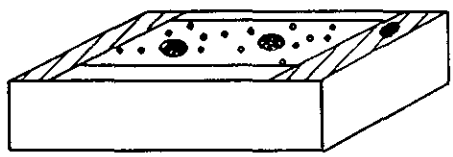
FIGURE 11. Cracks, chip outs or voids.

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REJECT: VISIBLE TERMINATION MATERIAL SPATTERED THROUGHOUT ACTIVE RESISTOR AREA.

FIGURE 12. Termination material.



REJECT: VISIBLE IMBEDDED FOREIGN PARTICLES IN EITHER THE RESISTOR FILM OR TERMINATION AREA. FOREIGN PARTICLES ARE THOSE THAT ARE NOT NORMAL TO RESISTOR DESIGN OR PROCESS.

FIGURE 13. Imbedded particles.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Reliability assurance program. A reliability assurance program shall be established and maintained in accordance with MIL-STD-790. Evidence of such compliance shall be verified by the qualifying activity of this specification as a prerequisite for qualification and continued qualification.

4.2 Classification of inspection. The examination and testing of resistors shall be classified as follows:

a. Qualification inspection (see 4.4).

b. Verification of Qualification (see 4.5).

c. Quality conformance inspection (see 4.6).

4.3 Inspections conditions and precautions.

4.3.1 Inspection conditions. Unless otherwise specified herein, all inspections shall be preformed in accordance with the test conditions specified in the "GENERAL REQUIREMENTS" of MIL-STD-202.

4.3.2 Precautions. Adequate precautions shall be taken during inspection to prevent condensation of moisture on resistors, except during the moisture resistance and temperature cycling tests.

4.4 Qualification inspection. Initial qualification shall be based upon the manufacturers' qualification to a minimum 'R' failure rate for all similar styles, of the equivalent MIL-R-55342 resistor being offered for listing on the qualified products list of this specification. Additionally, the manufacturer shall pass the qualifying activity audit (see appendix A.)

4.5 Verification of qualification. Every 6 months, the manufacturer shall compile a summary of the results of quality conformance inspections in the form of a verification of qualification report, and forward it to the qualifying activity within 30 days after the end of the reporting period as the basis of continued qualification approval. In addition, the manufacturer shall immediately notify the qualifying activity (within 15 days) whenever the data indicates that the manufacturer has failed to maintain his qualified product. Continuation shall be based on evidence that over the 6 month period the following has been met:

a. The manufacturer has not modified the design of the item.

b. Parts manufactured have met all specification requirements.

c. Lot rejection for group A inspections does not exceed 3 percent or one lot, whichever is greater.

4.6 Quality conformance inspection. Parts supplied to this specification shall have been manufactured on the same production line as the equivalent MIL-R-55342 resistor.

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4.6.1 Inspection of product for delivery. Inspection of product for delivery shall consist of group A inspection. Delivery of products which have passed group A shall not be delayed pending the results of group B inspection.

4.6.1.1 Production lot (group A). A production lot shall be of one design as specified in the baseline documentation. As a minimum requirement, the lot shall consist of resistors of the same style, characteristic, termination, resistance value, voltage rating and tolerance, and shall use substrates from a single lot and from a single deposition/metallization run. All resistors in the lot shall have been started, processed, assembled and tested as a group on the same manufacturing/test equipment, to the same baseline/revision, and identified with the same lot date code designation. Traceability of all raw materials used in the lot shall be maintained. In addition, the deposition/metallization shall be restricted to the following:

a. The substrate used shall be from a single lot.

b. The substrate produced shall be for a single resistor style.

c. The substrate shall be processed in a single continuous metallization run, on the same equipment.

d. The raw materials used in the metallization process shall be from the same raw material lots.

4.6.1.1.1 Rework. Rework is not allowed on any operation except marking.

4.6.1.2 Group A, subgroups 1, 2, and 3 inspection. Group A, subgroups 1, 2, and 3 inspection shall consist of the inspections specified in table VIII and shall be made on the same set of sample units, in the order shown.

4.6.1.2.1 Subgroup 1 tests. Subgroup 1 tests shall be performed on a production lot basis on 100 percent of the product supplied under this specification. Resistors that are out of resistance tolerance, or which experience a change in resistance greater than that permitted for the tests of this subgroup shall be removed from the lot. Lots having more than 3 percent total rejects, due to exceeding the specified resistance tolerance change limit, shall not be furnished on contracts.

4.6.1.2.2 Subgroup 2 tests.

4.6.1.2.2.1 Sampling plan. Thirteen samples shall be selected randomly from each inspection lot and subjected to the subgroup 4 solderability test. The manufacturer may use electrical rejects from the subgroup 1 screening tests for all or part of the samples to be used for solderability testing. If there are one or more defects, the lots shall be considered to have failed.

4.6.1.2.2.2 Rejected lots. Each production lot that was used to form the failed inspection lot shall be individually submitted to the solderability test as required in 4.7.10. An additional 13 samples from each production lot shall be tested. If the lot fails this solderability test, the lot may be reworked a second time and retested. If the lot fails the second rework, the lot shall be considered rejected and shall not be furnished against the requirements of this specification. Individual production lots that pass the solderability test are available for shipment.

4.6.1.2.3 Subgroup 3 tests. Subgroup 3 shall be performed on an inspection lot basis formed from one daily production. A sample of five parts shall then be randomly selected, if one or more defects are found, the lot shall be rejected and shall not be supplied to this specification.

4.6.1.2.4 Manufacturer's production inspection. If the manufacturer performs tests equal to or more stringent than those specified in subgroup 1, table VIII, as the final step of this production process, group A, subgroup 1 inspection may be waived and the data resulting from the manufacturer's production tests may be used instead. Authority to waive the subgroup 1 inspection shall be granted by the qualifying activity only. The following criteria must be complied with:

a. Tests conducted by the manufacturer during production shall be clearly identical to or more stringent than that specified for subgroup 1. Test conditions shall be equal to or more stringent than those specified for subgroup 1.

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b. Manufacturer subjects 100 percent of the product supplied under this specification to his production tests.

c. The parameters measured and the failure criteria shall be the same or more stringent than those specified herein.

d. The lot rejection criteria is the same or more stringent than that specified herein.

e. The manufacturer shall make available all information concerning the test procedures and instrumentation used in his production tests. This data shall be provided as part of the evaluation required for MIL-STD-790 and shall be documented in the baseline. The manufacturer shall also make available to the Government all records of all detail test data resulting from production tests.

f. Once approved the manufacturer shall not change the test procedures or criteria without prior notification and concurrence by the qualifying activity and subsequent change in the baseline.

TABLE VIII. Group A inspection.

Inspection	Requirement	Method	Sampling procedure
Subgroup 1			
Preprocessing visual	3.17 through 3.17.5		100 percent inspection
DC resistance	3.7	4.7.2	
Thermal shock	3.8	4.7.3	
Power conditioning	3.11	4.7.6	
DC resistance	3.7	4.7.2	
Visual and mechanical	3.17 through 3.17.5		
Subgroup 2			
Solderability	3.15	4.7.10	13 samples, 0 failures
Subgroup 3			
Destructive physical analysis (DPA)	3.18	4.7.12	5 samples, 0 failures
Subgroup 4			
Resistance to bonding exposure	3.12	4.7.7	13 samples, 0 failures
Resistance temperature characteristic	3.14	4.7.9	
Low temperature operation	3.9	4.7.4	
Short time overload	3.10	4.7.5	
Subgroup 5			
Adhesion	3.16	4.7.11	13 samples, 0 failures
Subgroup 6			
Minimum marking legibility	3.20.1 or 3.20.2	4.7.1.1 or 4.7.1.2	8 samples, 0 failures

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4.6.1.2.5 Group A, subgroups 4, 5 and 6 inspection. Group A, subgroups 4, 5 and 6 inspection shall consist of the inspections specified in table VIII in the order shown, and shall be performed on sample units which have been subjected to and passed the group A, subgroups 1, 2, and 3 inspection and shall represent one weekly production.

4.6.1.2.5.1 Sampling plan.

4.6.1.2.5.1.1 Subgroup 4. A sample of 13 parts shall then be randomly selected, if one or more defects are found, the lot shall be rescreened and defects removed. A new sample of 13 parts shall then randomly be selected, if one or more defects are found in this second sample, the lot shall be rejected and shall not be supplied to this specification.

4.6.1.2.5.1.2 Subgroup 5. A sample of 13 parts shall then be randomly selected, if one or more defects are found, the lot shall be rescreened and defects removed. A new sample of 13 parts shall then randomly be selected, if one or more defects are found in this second sample, the lot shall be rejected and shall not be supplied to this specification.

4.6.1.2.5.1.3 Subgroup 6. A sample of 8 parts shall then be randomly selected, from each inspection lot and subjected to the subgroup 6 tests. The manufacturer may use electrical rejects from subgroup 1 screening tests for all or part of the samples to be used for the subgroup 6 testing. If there are one or more defects, the lot shall be considered to have failed.

4.6.1.2.5.2 Disposition of sample units. Sample units which have been subjected to group A, subgroups 4 and 5 inspection shall not be delivered on the contract or purchase order.

4.6.2 Periodic inspection.

4.6.2.1 Group B inspection. Group B inspection shall consist of the tests specified in table IX, in the order shown. Group B inspection shall be made on sample units selected from lots which have passed group A inspection. Group B samples shall be representative of production. The allowable number of defective shall be as indicated in table IX. The assigned periodic tests shall be initiated within 30 days of the specified periodic inspection.

4.6.2.1.1 Sampling plan.

4.6.2.1.1.1 Monthly (subgroup 1). Every month, 10 units of each style, characteristic, and termination material and of any resistance value shall be subjected to the monthly tests specified in table IX.

4.6.2.1.1.2 Monthly (subgroup 2). Every month, 10 units of each style, characteristic, and termination material and of the highest resistance value produced during the month shall be subjected to the monthly tests specified in table IX.

4.6.2.1.2 Disposition of samples. Sample units which have been subjected to group B inspection shall not be delivered on the contract or purchase order.

4.6.2.1.3 Noncompliance. If a sample fails to pass group B inspection, the contractor shall take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which were manufactured under essentially the same conditions, with essentially the same materials, processes, etc., and which are considered subject to the same failure. Acceptance of the product shall be discontinued until corrective action, acceptable to the Government, has been taken. After the corrective action has been taken, group B inspection shall be repeated on additional sample units (all inspection, or the inspection which the original sample failed, at the option of the Government). Group A inspections may be reinstituted; however, final acceptance shall be withheld until group B reinspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure and corrective action taken shall be furnished to the cognizant inspection activity and the qualifying activity.

4.6.3 Inspection of packaging. Sample packages and packs and the inspection of the preservation packaging, packing, and marking for shipment and storage shall be in accordance with the requirements of MIL-R-39032.

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4.6.4 Retention of data and test samples. The manufacturer shall retain the following data and test samples for 12 years.

a. Group A samples.

b. DPA and solderability samples.

c. Lot travelers.

d. Lot test information.

e. Any rework data/samples, corrective actions, failure analysis, etc.

4.7 Methods of inspection.

4.7.1 Visual and mechanical inspection. Resistors shall be examined under 30X to 60X magnification, but in case of conflict, 30X will be the referee power unless otherwise specified to verify that the materials, design, construction, physical dimension and workmanship are in accordance with the applicable requirements (see 3.1, 3.3, 3.4, 3.4.1 through 3.4.3 inclusive, and 3.21).

4.7.1.1 Minimum marking.

4.7.1.1.1 Lasor marking. Resistors shall be coated with .005 inch minimum of silicone resin insulating compound, type SR of MIL-I-46058. After curing, coated resistors shall be examined for legibility under normal production room lighting by an inspector at 10X magnification.

4.7.1.1.2 Color dot/screen printing/stamp. Resistors shall be tested in accordance with method 215 of MIL-STD-202. After subjection to the test, any specified markings which are missing in whole or in part, faded, smeared, blurred, or shifted (dislodged) to the extent that they cannot be readily identified from a distance of at least 6 inches with normal room lighting without the aid of magnification or with a viewer having a magnification no greater than 3X shall constitute.

4.7.1.2 Chip handling procedures. When specified herein, the chip resistor shall be test mounted on a test substrate as described in 4.7.1.3. For those test procedures where mounting requirements are unspecified, the chip resistors may be test mounted using pressure type contacts.

TABLE IX. Group B inspection.

Inspection	Requirement paragraph	Method paragraph	Number of samples	Number of defects allowed
Subgroup 1				
Thermal shock (100 cycles)	3.8	4.7.3	10	0
Subgroup 2				
Moisture resistance	3.13	4.7.8	10	0

4.7.1.3 Specified mounting. When specified in the test procedure, the chip resistor shall be mounted on a suitable test substrate which shall be a minimum of 95 percent alumina (see figure 14). The test substrate material shall be such that it shall not be the cause of, nor contribute to any failure of a chip resistor in any of the tests for which it may be used. The test substrate shall be prepared with metallized surface land areas of proper spacing so that a test substrate surface area of at least four times the resistor chip surface is provided for each resistor chip mounted. The metallized surface land areas shall be designed in pattern to accommodate a number of resistor chips. The metallization material shall be compatible with bonding technique to be employed and the material used on the chip termination. The method of chip mounting for the termination materials shall be as follows:

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a. All terminations (except P and W). All chip resistors (except P and W) terminations materials shall be mounted on the test substrate by soldering the chip termination directly to the test substrate metallized land areas in the following manner:

(1) Solder paste equivalent to SN62 type "R" or "RMA" solder or liquid type "R" or "RMA" flux in accordance with QQ-S-571 shall be applied to the terminations of each chip code letters D, T, C, U, R, S, and B (see 1.2.4).

(2) All chips shall then be placed across the metallized land areas of the test substrate with contact between the chip terminations and substrate land areas only. The use of adhesive to keep chips in place during mounting operation is allowable.

(3) The test substrate with all chips in position shall then be placed on a suitable heat transfer unit (molten solder, hot plate, tunnel oven, etc.) with the temperature maintained at 260°C for 10 ±2 seconds except chips with termination B shall be maintained at 260°C for 60 ±5 seconds.

b. Termination material "W" and "P" (weldable). Chip resistors with weldable terminations shall be mechanically attached to the test substrate and electrically connected by thermocompression bonding flying lead interconnections between the chip termination and the test substrate metallized land areas. The interconnecting lead shall be .001 inch in diameter gold wire. The chip shall be mechanically mounted by any procedure which will not be the cause of or contribute to any failures of the chip resistor in any test.

c. At the manufacturer's option 4.7.1.3a(1) or 4.7.1.3b may be used.

d. Alternative mounting for terminations D, C, T, and U. As an alternative to 4.7.1.3a(1), chip resistors can be mechanically attached to the test substrate with type Z epoxy in accordance MMM-A-1931. The epoxy shall be cured at a minimum temperature of 150°C. The minimum curing time shall be 30 minutes.

4.7.2 DC resistance (see 3.7). The dc resistance shall be measured in accordance with method 303 of MIL-STD-202. The following details and exceptions shall apply:

a. Measuring apparatus: Different types of measuring test equipment (multimeters, bridges, or equivalent) are permitted to be used on the initial and final readings of this test, provided the equipment is the same style, model, or if it can be shown that the performance of the equipment is equivalent or better.

b. Test voltage: Measurements of resistance shall be made by using the test voltages up to the maximum values specified in table X. The test voltage chosen, whether it be the maximum or a lower voltage which would still provide the sensitivity required, shall be applied across the terminals of the resistor. This same voltage shall be used whenever a subsequent resistance measurement is made.

c. Temperature: The temperature at which subsequent and final resistance measurements are made in each test shall be within ±2°C of the temperature at which the initial resistance measurement was made.

TABLE X. Standard dc resistance test voltages.

Nominal resistance	Maximum test voltage			
	10 to 24 milliwatts	25 to 99 milliwatts	100 to 225 milliwatts	226 to 1000 milliwatts
Ohms	Volts	Volts	Volts	Volts
5.60 to 9.88	0.2	0.3	0.7	1.0
10.0 to 98.8	0.3	0.3	0.3	1.0
100 to 988	1.0	1.0	1.0	3.0
1,000 to 9,880	3.0	3.0	3.0	10.0
10,000 to 98,000	10.0	10.0	10.0	30.0
100,000 or higher	30.0	30.0	30.0	100.0

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4.7.3 Thermal shock (see 3.8). Resistors shall be tested in accordance with method 107 of MIL-STD-202. The following details and exceptions shall apply:

a. Mounting: Resistors shall not be mounted. They shall be placed in containers which allow circulation of air to each resistor.

b. Measurements before cycling:

(1) Group A inspection: DC resistance shall be measured as specified in 4.7.2, except at 25°C ±5°C. The dc resistance shall be within the specified resistance tolerance of the nominal resistance value.

(2) Group B test: DC resistance shall be measured as specified in 4.7.2.

c. Test condition F, except temperratures +150°C +10°C, -0°C and -65°C +0°C, -10°C.

d. Measurement after cycling:

(1) Group A inspection: After stabilization at room temperature, the dc resistance shall again be measured as specified 4.7.2, except at 25°C ±5°C.

(2) Group B test: After stabilization at room temperature, the dc resistance shall be measured as specified in 4.7.2. Following the test, the resistors shall be examined for evidence of mechanical damage.

4.7.4 Low temperature operation (see 3.9). A dc resistance shall be measured made as specified in 4.7.2, the resistor chips shall be mounted on a substrate a specified in 4.7.1.3. The resistors shall be placed in a cold chamber maintained at -65°C +0°C, -5°C. After 1 hour of stabilization at this temperature, full rated continuous working voltage, as specified in 3.7, shall be applied for 45 +5, -0 minutes. Fifteen +5, -0 minutes after removal of the voltage, the resistor shall be removed from the chamber and maintained at a temperature of 25°C ±5°C for a period of approximately 24 hours; the dc resistance shall then be measured as specified in 4.7.2. Resistors shall also be examined for evidence of mechanical damage.

4.7.5 Short time overload (see 3.10). The dc resistance shall be measured as specified in 4.7.2, on chip resistors mounted on a substrate as specified in 4.7.1.3. A dc test potential, 2.5 times the rated continuous working voltage but not exceeding twice the maximum voltage (see 3.1) shall be applied for 5 seconds to the resistor under the following conditions:

a. Resistors on substrate mounted so that larger space of the resistor is on a horizontal plane.

b. In still air, with no circulation other than that created by the heat of the resistors being operated.

Thirty +15, -0 minutes after removal of the test potential, the dc resistance shall again be measured as specified in 4.7.2. Resistors shall then be examined for evidence of arching, burning, and charring.

4.7.6 Power conditioning (see 3.11). A chip resistor shall be tested in accordance with method 108 of MIL-STD-202. The following details and exceptions shall apply:

a. Test temperature and tolerance: 70°C +5°C, -0°C.

b. Initial measurements: Measurements may be made inside or outside the chamber.

(1) Inside chamber: When measurements are to be made inside the chamber, the initial dc resistance shall be measured, at the applicable test temperature, after temperature stabilization, and within 8 hours of exposure of the chip resistors to the test temperature. This initial measurement shall be used as the reference temperature for all subsequent measurements under the same condition.

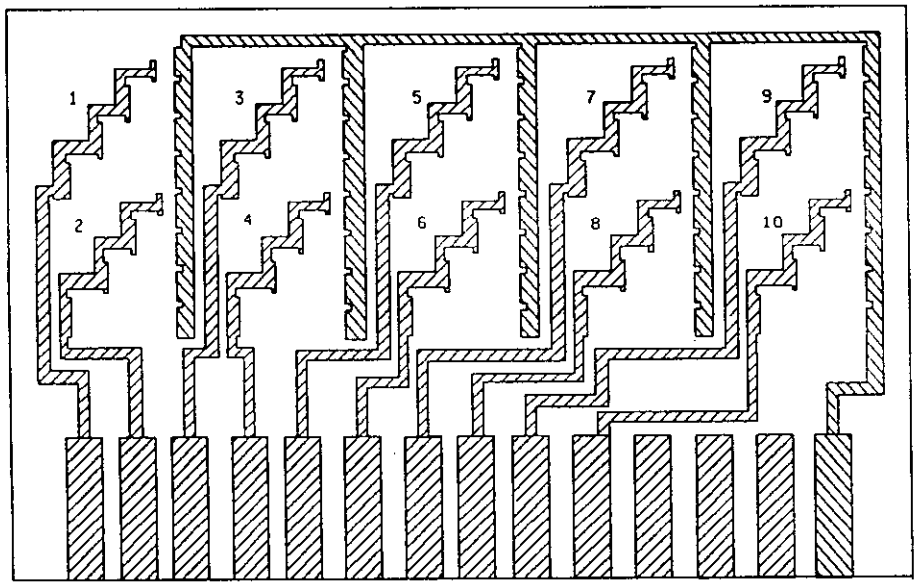
(2) Outside chamber: When measurements are to be made outside the chamber, the initial dc resistance shall be measured at the room temperature. This initial measurement shall be used as the reference temperature for all subsequent measurements under the same condition.

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c. Operating condition: Rated dc continuous working voltage, filtered or nonfiltered full-wave rectified ac voltage shall be applied to the chip resistor intermittently, 1.5 hours on and 0.5 hours off, for 100 ±4 hours and at the test temperature. During the "on" cycle, the voltage shall be regulated and controlled to maintain ±5 percent of the rated continuous working voltage. Unless otherwise specified (see 3.1), power applied shall be 1.5 times rated power.

d. Measurements after test: Following a minimum 0.5 hour stabilization period, dc resistance shall be measured as specified in 4.7.2.

e. Inspection after test: Chip resistors shall be inspected for evidence of mechanical damage.



NOTES:
1. Suggested substrate dimensions: 1.000 inch (25.40 mm) wide x 1.500 inches (38.10 mm) long x .025 inch (0.64 mm) thick.
2. The contacts fit a printed circuit board connector with a .100 inch (2.54 mm) contact center.
3. Short lead wires may be soldered on and will then fit a microcircuit connector. Long test lead wires may be soldered on for direct measurement.
4. Small grid squares are .010 inch (0.25 mm).

FIGURE 14. Suggested test substrate.

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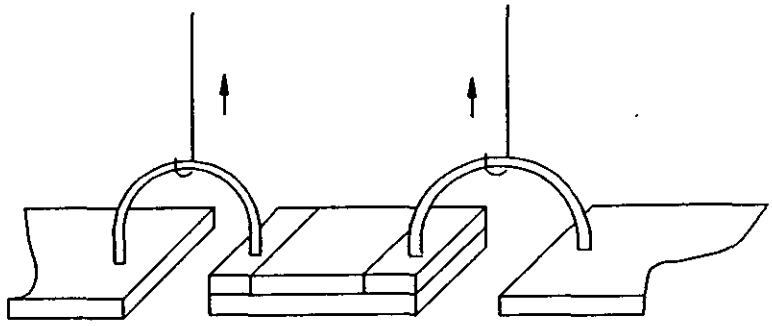


FIGURE 15. Weldable style units.

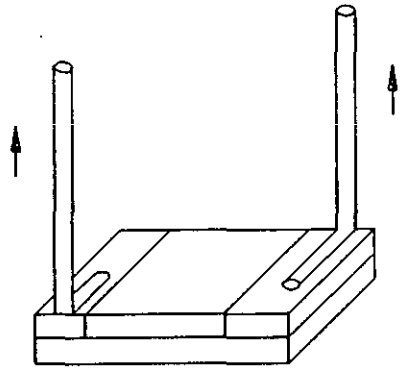


FIGURE 16. Solderable style units.

4.7.7 Resistance to bonding exposure (see 3.12). The dc resistance of the chip resistor shall be measured as specified in 4.7.2. The chip resistors shall then be mounted on a test substrate in accordance with 4.7.1.3. The test substrate with resistors mounted shall then be stabilized at 25°C ±3 °C after which the dc resistance shall again be measured as specified in 4.7.2. The change in resistance shall not exceed the value specified in 3.12.

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4.7.8 Moisture resistance (see 3.13). Resistors shall be tested in accordance with method 106 of MIL-STD-202. The following details and exceptions shall apply:

a. Mounting: Chip resistors shall be mounted on a test substrate as specified in 4.7.1.3.

b. Initial measurement: Immediately following the initial conditioning period, dc resistance of each chip on the test substrate shall be measured as specified in 4.7.2.

c. Polarization and loading voltages: The loading voltage shall be equal to 10 percent rated power as shown in the following example.

For R = 1 MΩ
V = 50 volts

$P = V^2/R = 50^2/1\ M\Omega = .0025\ w$

10% rated power = .00025 w

$V = \sqrt{PR} = \sqrt{.00025 \times 1\ M\Omega} \approx 15.8$ volts

d. Subcycle: Step 7b shall not be applicable. Step 7a shall again be performed during any five of the first nine cycles only.

e. Measurements at high humidity: None.

f. Final measurements: Upon completion of step 6 of the final cycle, the resistors shall be removed from the chamber and air dried for 30 ±15 minutes. The dc resistance shall then be measured in accordance with 4.7.2.

g. Examination after test: Resistors shall be examined for evidence of electrical and mechanical damage.

4.7.9 Resistance-temperature characteristic (see 3.14). Resistors shall be tested in accordance with method 304 of MIL-STD-202. The following details and exceptions shall apply:

a. Reference temperature: Room ambient temperature.

b. Test temperature: In accordance with table XI.

c. Accuracy of temperature measurement: Resistors shall be maintained for 30 to 45 minutes with 1°C at each of the test temperatures listed in table XI. This tolerance shall be maintained on the established test temperatures.

d. Mounting: Resistors shall be mounted on a substrate as specified in 4.7.1.3.

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TABLE XI. Resistance temperature characteristic.

Sequence	Temperature
	Group B acceptance inspection 1/
	°C
1.	Room temperature 2/
2.	-55 ±3
3.	Room temperature 2/
4.	125 ±3
5.	
6.	

1/ At the option of the manufacturer, the reverse sequence may be as follow:

2/ This temperature shall be considered the referee temperature for each of the succeeding temperatures.

1	Room temperature 2/
2	+125 ±3
3	Room temperature 2/
4	-55 ±3

As an option, the manufacturer may develop a control procedure for minimum stabilization times required for RTC testing. This procedure shall detail specific time elements required to reach thermal and electrical stability for specific part designs. Documentation of this procedure and supporting data shall be submitted to the qualifying activity for approval. Upon approval by the qualifying activity, the manufacturer may use this procedure for minimum stabilization times for RTC testing.

4.7.10 Solderability (see 3.15) applicable to B, G, and S terminations. Resistors shall be tested in accordance with MIL-STD-202, method 208. Both end terminations shall be immersed. The following details and exceptions shall apply:

a. Apparatus:

(1) A solder pot capable of keeping the solder temperature to ±5°C of set point.

(2) A device to keep the specimen submerged in solder for the required time.

(3) Optical equipment capable of 30X magnification.

(4) A laboratory beaker that is capable of:

(a) Supporting a nonmetallic device.

(b) Maintaining the test specimens at a distance of 1.5 to 2.5 inches from the surface of the boiling distilled water while, at the same time exposing them to full flow of steam generated.

b. The requirement for standard solderable wire application shall not apply.

c. The immersion and emersion rates shall not apply.

d. The specimen shall be totally submerged for at least 5 seconds.

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4.7.11 Adhesion (see 3.16). The resistor chips shall be prepared as specified in 4.7.11.1, 4.7.11.2 and 4.7.11.3 as applicable. A pull of 4 grams shall be applied to the weldable style units ("W" and "P") and 15 grams to all solderable, solderable-weldable units and bondable units. The force shall be applied at a 90 degree angle to the surface of the chip one lead at a time for 30 seconds each. The chip ceramic shall be held by means of tweezers or other suitable holding device. The resistor chips shall be examined for evidence of mechanical damage.

4.7.11.1 Weldable style units.

a. Method of mounting. Sample units shall be mounted on a test substrate as specified in 4.7.1.3.

b. Force. Force shall be applied by inserting a hook under the lead wire approximately in the center of the wire as shown on figure 15.

4.7.11.2 Solderable or bondable units. A 3 inch length of solid AWG No. 26 to No. 34 wire shall be soldered to the chip terminations as shown on figure 16.

4.7.11.3 Bondable units. A 3 inch length of solid AWG No. 26 to No. 34 wire shall be epoxy bonded to the chip terminations as shown on figure 16. The epoxy shall be type 2 in accordance with MMM-A-1931.

4.7.12 Destructive physical analysis (see 3.18). Resistors shall be examined as specified in appendix B of this document.

4.7.13 Outgassing (see 3.19). Resistors shall be tested in accordance with ASTM E595. The following details and exceptions shall apply. Samples of any materials used shall be cut into pieces having .38 inch (9.6 mm) maximum dimension. A sample of 100 to 300 milligrams of material shall be weighed and then placed in a vacuum chamber under the following conditions:

Pressure----------	10^{-6} torr or less.
Temperature of specimen----	Maximum operating temperature of device.
Exposure time----------	24 hours.

During this exposure, a collector plate, 9.7 inches (246.4 mm) in diameter, shall be held at 25°C ±1°C near the sample. Immediately upon removal of the specimen and collector plate from the chamber, they shall be weighed. The Total Mass Loss (TML) is sample mass before exposure minus the sample mass after exposure divided by the initial sample mass (expressed in percent change). The Volatile Condensable Material (VCM) is the increase in the weight of the collector plate divided by the mass of sample (expressed in percent).

5. PACKAGING

5.1 Packaging. The requirements for packaging shall be in accordance with MIL-R-39032.

6. NOTES

(This section contains information of general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Chip resistors are intended to be used in thick or thin film hybrid circuits where micro-circuitry is intended and in surface mounting applications.

6.2 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Title, number, and date of the applicable detail specification, and the complete PIN number (see 3.1 and 1.2.1).

c. Termination area configuration.

d. Substrate bonding (see 3.5.3).

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6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time set for opening of bids, qualified for inclusion in the applicable qualified products list whether or not such products have been so listed by that date. The attention of the manufacturers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the qualified products list is the Air Force Space and Missile Systems Center; however, information pertaining to qualification of products may be obtained from the Defense Electronics Supply Center (DESC-E), 1507 Wilmington Pike, Dayton, OH 45444-5278.

6.4 Application notes.

6.4.1 Mounting of resistors. Resistors may be mounted individually on a substrate, usually 95 percent alumina, and connected to conductor areas by means of solder paste, conductive cement or wire bonding, in some preform applications, solder preforms or solder paste can be used. Users are cautioned that the substrate and chip component must have compatible temperature coefficients of expansion to reduce the strain level imposed on the solder connections. They may also be directly connected to other components on the substrate by means of wire bonding, using the substrate as a base or carrier for the resistor.

6.4.2 Stacking of resistors. Stacking is in general discouraged since experience indicates that failure may occur due to electrolytic action in the bonding adhesive. This action occurs from the increase in temperature caused by stacking in conjunction with operating voltage across the resistor. The bonding adhesive at the resistor chip interface chemically reacts with the active resistor material changing the noise and resistance values and eventually the resistor will open. When stacking resistors, care should be taken to compensate for the lower heat dissipation capabilities by derating the wattage rating.

6.4.3 Close tolerance resistors. Close tolerance resistors (i.e., ±0.1 percent) should be mounted on a substrate by a method which produces the least heating effect over a short period of time to avoid permanent changes in resistance.

6.4.4 Resistor film orientation. It is recommended that the film side of resistors be mounted up when they are placed onto printed circuit boards. This is due to the effect of thermal stressing on epoxies used to hold them in place when mounted on the bottom of printed circuit boards. Resistor films can be torn from the substrate during thermal cycling, substantially altering the value of the resistor.

6.4.5 Tin plated finishes. *Tin plating is prohibited (see 3.4.2.2) since it may result in tin whisker* growth. Tin whisker growth could adversely affect the operation of electronic equipment systems. For additional information on this matter refer to ASTM B545 (Standard Specification for Electrodeposited Coating of Tin).

6.5 Noise level test. This specification does not include a noise test, but if the user requires a noise test, it shall be performed in accordance with method 308 of *MIL-STD-202. Excess noise has been reported in* resistors made using copper conductor films.

6.6 Electrostatic discharge effects. Under relative low humidity conditions, some types of film resistors, particularly those with small dimensions and high sheet resistivity materials, are prone to sudden significant changes in resistance (usually reductions in value) and to changes in temperature coefficient of resistance as a result of discharge of static charges built upon associated objects during handling, packaging, or shipment. Substitution of more suitable implements and materials can help minimize this problem. For example, use of cotton gloves, static eliminator devices, air humidifiers, and operator and work bench grounding systems can reduce static buildup during handling. Means of alleviating static problems during shipment include elimination of loose packaging of resistors and the use of metal foil and anti-static (partly conducting) plastic packaging materials.

6.7 Subject term (key word) listing.

Bondable
Solderable
Weldable

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APPENDIX A

PROCEDURES FOR THE PRODUCT AND QUALITY AUDIT

10. SCOPE

10.1 Scope. The appendix provides procedures to the Product and Quality Audit team for the auditing of manufacturers supplying to this specification for certification of manufacturing facilities. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

20. APPLICABLE DOCUMENTS. This section is not applicable to this appendix.

30. INSTRUCTIONS

30.1 Baseline document. The manufacturer shall complete the baseline document and submit the required information to the qualifying activity prior to scheduling the Product and Quality Audit. It is not intended that the manufacturer submit proprietary information. The baseline document shall contain the following information, as a minimum:

a. Detailed flow charts indicating the following:

(1) All significant process steps and measurement and inspection points by descriptive name and document number.

(2) All raw material and interim and finished part quality control points by descriptive name, document number, and location in the processing sequence. Documents shall include forms and travelers.

(3) All raw material incoming inspection, storage, and processing steps by descriptive name, document number, and location in the processing sequence. Documents shall include forms and travelers.

(4) The identification of all critical (significant to end product properties) manufacturing and inspection steps as determined by the manufacturer in conjunction with the audit team at the time of baseline document approval.

b. A listing of the documents identified in the flow charts identifying current revision letter, revision date and classification (i.e., critical, proprietary).

c. Blank copies of all current forms and travelers cited in the flow charts.

d. Copies of quality control documents detailing sampling plans used in inspections cited on flow charts and the quality control inspector training.

e. Procedures to control documentation, documentation changes, and distribution of the documents or document changes to all required personnel.

f. Procedures for raw material control for procurement, incoming inspection, storage, withdrawal from inventory, and shelf life control.

g. Details of part design delineating dimensions, tolerance, materials, and internal part numbers.

h. Procedures for controlling computer software including basic flow chart for program logic, current revision numbers, program access controls, and storage of part revisions of programs.

i. A complete list of process and test equipment utilized.

j. Organizations chart of key personnel and their relations.

k. List of raw materials and approved/qualified suppliers.

l. List of approved outside test and service laboratories used.

m. Procedures for statistical process control and failure analysis.

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30.1.1 Changes to baseline. The audit team shall compare the latest baseline document to the previous baseline document to assure all changes have been submitted to the qualifying activity and have been approved.

a. Minor changes. Any subsequent departure from the approved flow chart (i.e., additions or deletions) or changes to the revision status of those processes or inspections are allowed, provided they do not violate the changes prohibited by Section b (major changes). These minor changes shall be described in general terms to the audit team during subsequent audits. This action shall result in an update of the baseline document.

b. Major changes. No changes (i.e., substitutions, additions, deletions, or modifications) shall be allowed in the following areas without prior review and approval of the qualifying activity. In these, and other areas deemed major by the audit team at the time of baseline approval, partial or total product requalification may be required as determined by the qualifying activity.

(1) Significant raw materials.

(2) Qualified sources for significant raw materials.

(3) Substrate preparation.

(4) Procedure for metallization or deposition.

(5) Element stabilization.

(6) Trimming or resistance adjustment.

(7) Termination preparation, application, and sintering.

(8) Termination plating and cleaning.

(9) Coating process.

(10) Encapsulation or sealing process.

(11) Group A or B inspections.

30.2 Audits. An initial audit and subsequent periodic audits (every year or as deemed necessary) shall be performed by the qualifying activity. The audit shall include, but not be limited to, all of the items on the checklist (plus MIL-STD-790) and any GIDEP ALERTS issued against the manufacturer at the facility of interest. The results of any additional items that are assessed shall be added to the checklist for record purposes (see table XII).

30.2.1 Duties of the audit team.

a. The audit team shall review the manufacturer's facility and the proposed baseline documentation for adequacy, accuracy and completeness during the audit of the manufacturer. The proposed baseline documentation should be used as a baseline/reference during the course of the audit.

b. The audit team shall verify that the test measuring procedures and equipment are sufficiently accurate for the required test.

c. The audit team shall typically sample check 12 months of records for the items being certified. In addition, the location of the equipment and the thoroughness and adequacy of the operating procedures and instructions shall be checked. Whenever practical, operations utilizing critical equipment and procedures shall be witnessed.

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APPENDIX A

TABLE XII. Checklist.

Check - item	Inspection	Satisfactory
a. Verification of MIL-STD-790 requirements.		
b. Flow chart. (1) Key processes identified with revision. (2) Key inspection points identified with revision. (3) Quality control points identified with revision. (4) Key raw material control documents identified with revision. (5) Critical steps identified. (6) Nonproprietary documents reveiwed. (7) Test and measurement points identified with revisions.		
c. Baseline: Shall be describe, as a minimum, the following: (1) Resistor design by part number indicating the following with necessary tolerance defined: (a) Termination metallization (b) Body length (c) Body width (2) Documents listing suppliers qualified for all materials.		
d. Lot control plan (see 4.6.1.1).		
e. Traceability plan.		
f. Trainig plan for all personnel.		

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APPENDIX B

PROCEDURES FOR DESTRUCTIVE PHYSICAL ANALYSIS (DPA)

10. SCOPE

10.1 Scope. This appendix provides the methods and detailed requirements for DPA of chip film resistors. This appendix is a mandatory part of the specification contained herein and is intended for compliance.

20. APPLICABLE DOCUMENTS. This section is not applicable to this appendix.

30. EQUIPMENT AND PROCEDURES

30.1 Equipment.

30.1.1 Optical equipment. A binocular microscope with a minimum capability of 100X magnification and an integral light source or fiber optic light ring shall be used.

30.2 Resistor, fixed, film, style RSM0502, 0505, 1005, 1505, 2208, 0705, 1206, 2010, 2512, and 1010.

30.2.1 Method.

30.2.2 Visual inspection. Examine resistors at 30X minimum magnification for the following:

a. Cracked or chipped parts.

b. Foreign material or discoloration on the substrate, resistive film, or terminal bands.

c. Lifting, blistering, voiding, or peeling of the resistive film, terminal bands, or protective coating (when applied).

d. Any one rejection mode described on figure 17 of this appendix.

40. DESTRUCTIVE PHYSICAL ANALYSIS

40.1 Data records. DPA findings that deviate from configuration and other requirements shall be documented as defects.

40.2 Evaluation criteria. When the DPA is conducted as a lot conformance test, the production lot shall be rejected if one or more of the DPA samples exhibit any of the defects listed below. All exposed inner surfaces of each resistor shall be examined at 30X minimum magnification for the following characteristics:

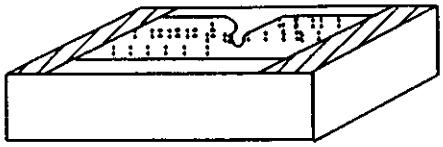
a. Cracked or chipped parts.

b. Foreign material or discoloration on the substrate, resistive film, or terminal bands

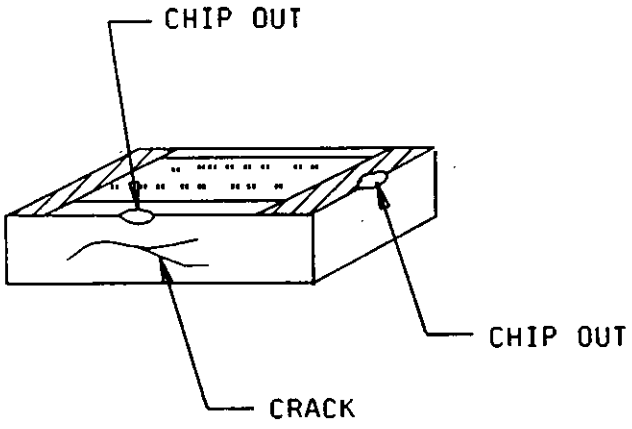
c. Lifting, blistering, voiding, or peeling of the resistive film, terminal bands, or protective coating (when applied).

d. Any one rejection mode described on figure 17 of this appendix.

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REJECT: VOID, NICK, BUBBLE OR CUT IN ACTIVE RESISTOR AREA THAT IS GREATER THAN 25 PERCENT OF THE RESISTOR WIDTH, EXCEPT CALIBRATION CUT. THE USABLE RESISTANCE WIDTH SHALL NOT BE LESS THAN 60 PERCENT OF THE ACTIVE RESISTOR ELEMENT.



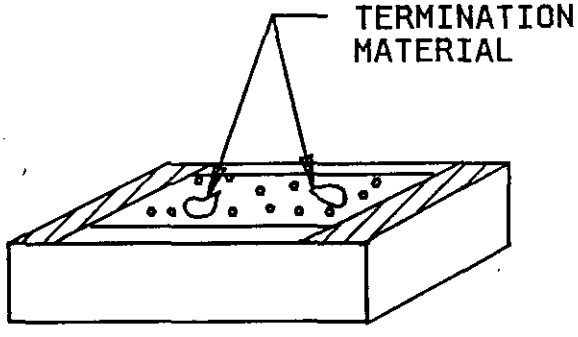
REJECT: ANY CRACK, CHIP OUT, OR VOID IN THE SUBSTRATE THAT IS GREATER THAN 0.08 MILLIMETERS (0.003 INCHES) IN ONE DIMENSION.

FIGURE 17. Chip resistor visual examination for all styles.

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REJECT: EXCESSIVE BUILD UP OF TERMINATION THAT IS GREATER THAN .003 INCH (0.08mm) AND .008 INCH (0.20mm) FOR SOLDERABLE AND WELDABLE FINISHES, RESPECTIVELY.



REJECT: TERMINATION MATERIAL SPLATTERED ON THE ACTIVE RESISTOR AREA.

FIGURE 17. Chip resistor visual examination for all styles - Continued.

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REJECT: FOREIGN PARTICLES THAT ARE EMBEDDED IN EITHER THE RESISTIVE FILM OR TERMINATION BANDS. (NOTE: FOREIGN PARTICLES ARE THOSE OTHER THAN THE MATERIALS USED AS RESISTIVE FILM OR TERMINATION BAND.)

FIGURE 17. Chip resistor visual examination for all styles - Continued.

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 19
NASA - NA

Review activities:
Army - AR, AT, AV, ME
Navy - AS, CG, MC, OS
Air Force - 17, 85, 99
DLA - ES

Preparing activity
Air Force - 19

Agent
DLA - ES

(Project 5905-1259)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-R-83580	2. DOCUMENT DATE (YYMMDD)

3. DOCUMENT TITLE

RESISTORS, FIXED, FILM, CHIP, ESTABLISHED RELIABILITY, GENERAL SPECIFICATION FOR

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME SSD/SDMS P.O.C ALAN WANG	b. TELEPHONE (Include Area Code) (1) Commercial 213-643-1966 (2) AUTOVON 833-1966
c. ADDRESS (Include Zip Code) P.O. Box 92960 Los Angeles AFS Los Angeles, CA 9009-2960	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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